1 Socio-Pragmatic Architecture and System Component Specifications

1.1 Pragmatics, Level-k Reasoning, and Searlean Realities

To model social networks where agents think about other agents' thoughts, a custom Probabilistic Domain Specific Language (DSL) was developed. This language parses recursively nested mental states (`thinks`, `chooses`, `knows`, `observes`) to simulate Rational Speech Act (RSA) matrices and level-k strategic convergence (e.g., Schelling games, political and religious belief polarization networks).

To satisfy John Searle's philosophy of the social world, the language was expanded to Grammar 3, embedding a dedicated, constitutive status operator:

$$\mathcal{D}_{\text{inception}}\Big(\langle C,\, X,\, Y\rangle\Big) \implies (X \text{ counts as } Y \text{ in } C) \tag{1}$$

This allows the engine to instantiate institutional facts that cannot be measured via physical sensory inputs. The grammar cleanly separates illocutionary forces into five distinct categories: Assertives, Directives, Commissives, Expressives, and Declarations. This framework manages security perimeters, access control, and legal obligations containing strict temporal deadlines (t_{deadline}).

1.2 Specification and System Components

The neuro-symbolic system acts as a bounded, type-safe transducer. It inputs structural strings or queries and produces learned contexts, classifications, or deductive responses based on the available training material.

The system is powered by seven interlocking components:

1. **LLM as Co-Designer:** Acts as an automated reward stencil synthesizer and meta-level instructor.
2. **Python Orchestrator:** The master continuous execution plane and virtual machine runner.
3. **Forest of Decision Trees:** The core associative, hierarchical memory array for learning situational contexts.
4. **The Prolog Guardrail (`kb.pl`):** The formal logical attention filter ensuring type-safe branch splits.
5. **RL with DSLs:** Sequential decision-making engines optimizing policies over bounded continuous languages.
6. **Automated Algebraic Solver:** Symmetric Group (S_3) controllers providing invertible, loss-free data rollbacks.
7. **Graph Neural Network (GNN):** PyTorch Geometric modules processing large-scale empirical church network graphs.

2 Context Layer: Interlocking Decision Forests

2.1 Decision Trees as Associative Agent Memory

Decision trees function as the agent's structural memory matrix and are used directly for context classification. The forest learns complex relational concepts as presented by sparse multi-modal exemplars. The non-terminal internal nodes are derived dynamically from the exemplars, evaluating testing features or structured parthood (`part_of`) relationships.

The decision tree recursion is explicitly configured to support nested conceptual compositionality, enabling the system to learn high-level meta-concepts strictly in terms of simpler, underlying local sub-concepts. Concepts are formulated in LISP S-expressions and can be indefinitely nested. Terminal leaf nodes retain partial exemplar arrays and domain-specific solution techniques. There are multiple independent trees trained for each category of concepts, ensuring that computationally analogous scenarios converge cleanly onto the exact same leaf node.

2.2 `kb.pl` Logic as the Attention Guardrail

The Prolog logic layer operates as an explainable attention filter. If a decision tree attempts to shift its focus or generate a new, random feature split to minimize statistical error over noisy data, the `kb.pl` file audits that choice. It blocks the tree from chasing noise or hallucinating branches, verifying that variables are legally and structurally meaningful to the macro-institution.

3 Training and Inferencing Plane

3.1 Reduced-Vocabulary Intrusion Detection Gate

To protect the system from un-vetted adversarial payloads, the LLM provides all prompts in a strictly reduced vocabulary form to the neuro-symbolic intrusion detection system (IDS). This system was engineered in a previous development phase; it is maintained as a decoupled, upstream filter boundary to allow current research to focus exclusively on multi-agent system (MAS) consensus networks and parallel branching poset architectures without character cross-talk.

3.2 Inductive Ontology Replacement and Naming Realignment

Our architecture replaces manual ontology revision entirely by utilizing the unique mathematical properties of learned decision trees, rendering the shared strategy completely *inductively definable*. The 5-step merging procedure unrolls at the target agent as follows: (1) Logic Basis Consensus on the received `kb.pl` file; (2) Model Payload Buffering of the source tree; (3) Logical Interrogation and Repair by stripping type names to blind queries; (4) Inductive Retraining Pass over updated exemplars; and (5) Ontology Substitution for a conflict-free semantic merger.

3.3 Emergent MARL CSMA/CD and Power Grid Performance Curves

We evaluate decentralized Multi-Agent Reinforcement Learning (MARL) nodes cooperating to share grid infrastructure without top-down centralized scheduling. The deep reinforcement learning engines dynamically discover an emergent Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol. The aggregate load on the transmission line functions as the physical carrier channel. If multiple rival data center nodes inject compute workloads simultaneously, the resulting synchronous load episode is instantly captured via SCADA telemetry as a channel collision.

Following the structural network implementation principles first established in **Sherman & Gable (1983)**, the local MARL controllers respond to this collision detection signal by instantly aborting their active training matrices to maximize collective rewards, applying an exponential backoff delay to retry during an alternative, non-overlapping time step.

3.4 Besnard-Hunter Deductive Argumentation and Defeasible Self-Repair

When an incoming theory package from a source agent violates a global core invariant, classical First-Order Logic (FOL) collapses due to trivialization. To bypass this vulnerability, the architecture implements Philippe Besnard and Anthony Hunter's deductive argumentation framework, modeling interactions as structured tuples $\mathcal{A} = \langle \Phi, \alpha \rangle$, where Φ is the logically consistent support subset and α is the claim. When a logic collision occurs, the target agent constructs an active *Undercutter* counter-argument targeting the consistency of the source's support premises. The undercutter payload triggers an automated self-repair loop over the Model Context Protocol (MCP) tool layer, forcing the corrupted node to execute local back-propagation pruning and retrain its abstract tree topology without generating a global system halt.

3.5 Cross-Domain Isomorphism and Typological Analogies

We scale this architecture to support cross-domain reasoning by analogy, demonstrating that the structural form of an optimization problem applies with identical integrity across physical, social, and spiritual lattices. When the intensional type-label stripping operator purges localized name tags, disparate realities map to identical geometric paths within the decision forest. This allows the system to unify physical SCADA grid surges, John Searle's institutional status assignments, and scriptural-spiritual salvific extractions (such as mapping Judith's mountain crevice rescue directly to the non-convex trap coordinates of the Prodigal Son). God's perspective operates as the ultimate atemporal sovereign controller (W_2) at the apex root node ($\top$), breaking the closed syntax limits of local human observation planes to enforce absolute safe performance perimeters.

4 Conclusion and Future Horizons

We have presented a robust, unified Neuro-Symbolic AI architecture that successfully scales across continuous physical motion, recursive theory of mind lattices, and discrete Searle institutional speech acts. By grounding continuous neural transformations into explainable symbolic guardrails, the framework eliminates VRAM memory black-box opacity and prevents hallucinatory logic paths.

5 References

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